International Journal of Extreme Manufacturing

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To cite this article before publication: Zechao Han *et al* 2026 *Int. J. Extrem. Manuf.* in press https://doi.org/10.1088/2631-7990/ae83ec

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Journal:	*International Journal of Extreme Manufacturing*
Manuscript ID	IJEM-113274.R1
Manuscript Type:	Paper
Keywords:	Femtosecond laser, Micro/nanostructures, Pressure sensors, High sensitivity, Broad sensing range

Femtosecond laser-engraved ultra-broad-range pressure sensors with enhanced sensitivity

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Abstract: Achieving flexible pressure sensors that simultaneously combine ultra-high sensitivity, ultra-broad detection range, and low detection limit remains a major challenge because of the intrinsic trade-off between signal sensitivity and working span. Here, we demonstrate a femtosecond laser engraving approach that directly constructs programmable hybrid micro/nanostructures on metallic electrodes in a single, mask-free step. This top-down laser patterning process produces dual-scale micro/nanostructures, significantly expanding the interfacial coupling region and enhancing sensor performance. The resulting iontronic pressure sensors exhibit extreme capabilities, characterized by a maximum sensitivity reaching 6 414 kPa^{-1}, a wide sensing range up to 800 kPa, a low detection threshold of 3.4 Pa, a rapid signal response (~30 ms), and excellent cycling stability. For demonstration, the sensors were integrated into a wearable multi-octave glove system, and their robustness in multifunctional pressure recognition was validated. This work highlights the use of the femtosecond laser-based approach as an efficient and scalable manufacturing method for tunable, hybrid micro/nanostructures, opening new opportunities for advanced electronic skins, intelligent robotics, and human–machine interfaces.

Keywords: femtosecond laser, micro/nanostructures, pressure sensors, high sensitivity, broad sensing range

Introduction

With recent advances in artificial intelligence, big data, and robotics, pressure sensors—especially flexible types—have emerged as highly promising technologies for wearable electronics, electronic skin, and smart human–machine interfaces[1,2]. They are superior to other technologies because of their low cost, mechanical flexibility, simple fabrication processes, and superior integration potential[3,4]. In applications ranging from prosthetics and robotics to health monitoring[5,6] and virtual reality interfaces[7], sensors must reliably capture a broad spectrum of forces, from minuscule pressures like pulse waves and gentle touches (<10 Pa) to strong compressions in the hundreds of kilopascals. Simultaneously, these methods require extremely high sensitivity and fine pressure resolution to distinguish subtle force variations[8,9]. Accurately mimicking human tactile sensation, which spans approximately 0.1 kPa to more than 100 kPa, demands sensors that maintain high performance across this full range[10,11]. However, realizing all of these properties in a single device is extremely challenging[12]. Enhanced sensitivity often leads to early saturation at moderate pressures, whereas designs optimized for ultrabroad range tend to exhibit low sensitivity at light loads[13,14].

Among various sensing technologies, capacitive sensors have been widely used for constructing advanced pressure sensors owing to their simple device structure and reliable response to both static and dynamic pressures[15]. Conventional capacitive sensors often suffer from limited sensitivity and signal quality due to the intrinsically incompressible dielectric layer[16]. In this context, iontronic sensing, which relies on electric double layer (EDL) formation at the electrode–active layer interface, can achieve very high sensitivity through enhanced interfacial capacitance[8,17]. However, iontronic sensors also face a realistic tradeoff between high sensitivity and an extended detection range arising from the low compressibility and stiffness of microstructures with increasing pressure[5,13]. To address these challenges, attempts have been made to explore various structures to increase the specific contact area, including a graded intrafillable architecture (sensitivity of 220 kPa^{-1} over 360 kPa)[17] and printed interlocks (sensitivity reaching 49.1 kPa^{-1} within 485 kPa)[18]. While such strategies improve performance, the fabrication of microstructures relies on complex photolithography or transfer from a random template[19,20]. In addition, further increasing specific contact areas by fabricating hybrid micro/nanostructures on ionic layers is also challenging through these processes[15]. Recent advancements in laser-induced micro/nanostructuring have significantly propelled the development of intelligent sensing systems[21–24] and electronic skin technologies, underscoring the vast potential of these systems for enhancing flexible pressure sensors[25–27]. These developments inspire us to harness the inherent ability of the laser to produce high-precision micro/nanostructures efficiently and at scale for flexible pressure sensor fabrication[28,29].

In this work, we introduce a femtosecond laser-engraved hybrid micro/nanostructured pressure sensor that overcomes the aforementioned trade-offs. A single-step laser patterning process directly creates hierarchical features on the device surfaces, with micrometer-scale relief structures imparting mechanical compliance and compressibility, whereas laser-induced nanostructures dramatically increase the effective interfacial area as the pressure increases[30–32]. The geometry and height of each pyramid frame array can be individually tuned and optimized to achieve enhanced performance, leading to an ultra-broad working range extending to 800 kPa with an impressive sensitivity of 6 414 kPa^{-1}. The synergy between the programmed hybrid micro/nanostructure array also results in a minimal detectable pressure of 3.4 Pa and high-pressure resolution. Moreover, the sensor maintains stable performance over thousands of loading cycles, underscoring its mechanical robustness and reliability. Importantly, mask-free laser fabrication is rapid and scalable, enabling efficient production of large-area sensor arrays without complex lithography. We further applied an iontronic pressure sensor in multi-octave musical glove controlled by finger gestures and multi-level pressure on a broad scale. Finally, we demonstrate that such sensors can function as a capacitive electronic skin capable of providing tactile feedback for objects across a wide range of weights. Combined with its scalable fabrication capability, this study offers an effective route to fabricate highly sensitive and ultra-broad-range pressure sensors for next-generation electronic skins and smart intelligent human–machine systems.

Results and discussion

The microstructures and compressive conditions at the interfaces of iontronic devices are crucial in determining the overall sensing characteristics. Previous studies have demonstrated that well-designed microstructural geometries, which effectively enlarge the contact area at the interface, can modify and enhance both the sensitivity and the operational range of such devices[33,34]. Nonetheless, the structural features of active ionic layer achieved by conventional fabrication approaches are usually micrometer in size, and further decreasing the microstructures down to the nanoscale to increase the contact area further is quite challenging. In this study, a nanoscale iontronic interface was realized at micronized contact in sensors by simply applying femtosecond laser scanning on thin titanium foil. As femtosecond laser ablation is a top-down fabrication strategy capable of producing tunable micro/nanostructures on a wide variety of materials, it is particularly beneficial for capacitive iontronic pressure sensors to enhance sensing performance by enlarging the interfacial areas[35,36]. The high-intensity energy of ultrashort pulses can effectively remove the materials of the substrate, forming the ordered microscale frames. Owing to the ultra-high peak power input, a plasma plume containing abundant nanoparticles is generated and rapidly redeposited onto the cooler surfaces, producing stable nanoscale features on microstructures, as shown in **Figure 1(a)**[37]. This programmable, single-step laser patterning process thus enables mask-free fabrication of tunable hybrid micro/nanostructure arrays. As shown in **Figure 1(b)**, the laser-processed surfaces exhibit dual-scale structures: the microscale frames are generally pyramidal, with slightly curved sidewalls because of the Gaussian energy profile of the laser beam, whereas dense nanoclusters 100–700 nm in diameter (**Figure S1**) are anchored on the frame surfaces, substantially increasing the specific contact area and contributing to the extended sensing range.

Following laser fabrication, a thin gold film was deposited onto the electrode substrate to ensure good electrical conductivity (**Figure S2**). A composite ionic film composed of poly(vinyl alcohol) (PVA) and phosphoric acid (H_3PO_4) was sandwiched between two laser-engraved electrodes with micro/nano hybrid structures, and a spacer of annular polydimethylsiloxane (PDMS) was used to fix the gap and support the device (**Figure 1(c)**). Inserting an annular spacer between the upper electrode and the dielectric layer suppresses premature electrode–dielectric contact, thereby reducing the initial capacitance (C_0) and amplifying the relative capacitance variation ($\Delta C/C_0$) under pressure (**Figure S3**). **Figure S4** illustrates the assembly process of the iontronic pressure sensor. The entire device was finally encapsulated with two polyacrylate elastomers to ensure robust sensing performance under external force loading. Photographs of the structured electrodes and the hydrogel dielectric film are provided in **Figures S5 and S6**. The excellent performance of the laser-engraved iontronic sensor arises from the strong interfacial capacitance formed at the electrode–dielectric interface, known as the EDL capacitance. As illustrated in **Figure 1(d)**, under applied pressure and an external electric field, free electrons in the electrode layers attract oppositely charged mobile ions within the dielectric film. The gathering of countless electrons and ions at the interface forms nanoscale capacitors with extremely short charge separation distances, thereby generating an ultrahigh capacitance. Because the total capacitance is directly correlated with the effective interfacial area, the femtosecond laser-engraved hybrid structure significantly increases the surface area and thus enhances the overall sensing capability. In contrast, sensors without such structures significantly decrease the sensitivity, as displayed in **Figure S7**. An image of the encapsulated iontronic device is provided in **Figure 1(e)**. Owing to its high sensitivity, broad working regime, and ultralow detection limit, the device can be readily integrated into wearable human–machine interface systems such as the multioctave musical glove (**Figure 1(f)**).

As the interfacial configuration and sensing behavior of iontronic sensors are strongly dependent on their microstructural features, the geometry design and optimization of micro/nano hybrid structures are therefore of great significance (**Figure 2(a)**). To systematically elucidate how femtosecond laser-engraved micropyramid frames influence the sensor performance, we designed and fabricated arrays of different pyramidal structures. The width (D) and height (H) were controlled by the laser scanning pitch and laser input power, respectively. **Figure 2(b)** presents top-view scanning electron microscope (SEM) images of electrodes with different widths under a fixed height of 32 μm, while **Figure 2(c)** presents cross-sectional SEM images with different heights under a fixed width of 50 μm. **Figures S8–S15** provide additional SEM images from top, cross-sectional, and 45°-tilted views. It can be seen that both D and H can be easily adjusted by tuning the laser fabrication parameters. When D is much larger than the spot size of the femtosecond laser (~20 μm), most of the area of the top pyramid is flat, with only edge regions containing nanoclusters (**Figure S11**). With decreasing D, the area with nanostructures increases gradually. At D = 50 μm, most of the top surface of the pyramid is covered with nanoclusters (**Figure S9**). However, when D is further decreased to 20 μm, which is smaller than the spot size, the pyramid appears re-ablated and the effective area of the top surface decreases (**Figure S8**). In addition, by increasing the input power, the ablation depth (height of the pyramid) by femtosecond laser is continuously increased (**Figure 2(c)**).

We next characterize the sensing performance of the device with different values of D and H. For capacitive pressure sensors, the sensitivity (S) is defined as $S = \delta(\Delta C/C_0)/\delta P$, where δ, ΔC, C_0, and P denote the infinitesimal variation, the capacitance change, the initial capacitance, and the applied pressure, respectively. We first evaluate the effect of D while keeping $H = 32$ μm. As shown in **Figure 2(d)**, denser micropyramid arrays (i.e., smaller D) exhibit better sensitivity, linearity, and detection range due to greater contact area under compression. Sensors with sparser structures ($D = 100$ μm or 200 μm) show reduced sensitivity, and their normalized capacitance begins to plateau at approximately 200 kPa, reaching saturation near 800 kPa. Conversely, overly dense arrays ($D = 20$ μm) also underperform, with the signal saturating prematurely under high pressure. This is mainly because the scanning pitch is comparable to the laser spot diameter, causing thermal interference that unintentionally ablates sharp conical tips atop the micropyramids (**Figure S8**). These overly sharp structures are prone to deformation or collapse under loading and fail to support the dielectric layer. As a result, the dielectric fully intrudes into the narrow gaps, leaving little vertical depth for further compression or contact area expansion.

We also investigated the effect of H on sensing performance by setting $D = 50$ μm. As shown in **Figure 2(e)**, the results aligned with expectations: taller microstructures induce greater changes in the interfacial contact area under pressure, thereby enhancing both the sensitivity and working range. However, excessively tall structures ($H = 40$ μm) exhibit inferior performance. Although taller structures theoretically offer a larger surface area, the limited deformability of the ionic gel prevents it from fully infiltrating narrow grooves. Consequently, the effective interfacial contact area and the localized electric field enhancement are confined to the upper openings of the structures, leading to premature signal saturation and reduced sensitivity. These results underscore the importance of synergistically optimizing the width and height of microstructures to balance mechanical compliance and interfacial efficiency. A comprehensive summary of the effects of width and height is presented in **Figure 2(f)**, demonstrating that the micropyramids with moderate dimensions ($D = 50$ μm, $H = 32$ μm) used in this work achieve a favorable balance of high sensitivity, broad detection range, and good linearity.

To understand the role of micro/nanostructures, both experimental and simulation methods were employed. Experimental verification was carried out by comparing the performance of devices with different micro/nanostructure configurations. As shown in **Figure 2(g)**, the performance of devices lacking nanostructures and those with no micro/nanostructures is notably lower than that of devices with intact micro/nanostructures, confirming that the presence of micro/nanostructures plays a significant role in enhancing sensor performance. The optical images presented in **Figure S16** illustrate the gradual increase in contact area as the pressure increases, highlighting the relationship between the contact area and applied force. Finite element analysis (FEA) was subsequently conducted to further explore the underlying mechanisms. Owing to computational constraints, the nanostructures were intentionally exaggerated in scale and modeled as several semicircular protrusions of varying sizes on top of the microstructures. **Figures S17–S19** present detailed 2D and 3D simulation results, including comparisons of the stress distribution and contact region between electrodes with and without nanostructures. As shown in **Figure 2(h)**, the stress distribution is more uniform and concentrated near the nanostructured surface, indicating improved stress transfer and mechanical coupling with the dielectric layer. The ionic gel has deformed to closely conform around the nanostructures, suggesting a significantly enlarged effective area at the interface. Moreover, in **Figure 2(i),** the simulation results of $\Delta A/A_0$ for the nanostructured case exhibit a response trend similar to the experimental $\Delta C/C_0$ curve with pressure, suggesting that the significant performance improvement due to the micro/nanostructures mainly originates from the increased contact area at the interface, which amplifies the EDL effect.

As shown in **Figure 3(a)**, the normalized capacitance change ($\Delta C/C_0$) with respect to the applied pressure can be divided into three linear ranges. At zero pressure, the annular spacer separates the upper electrode from the dielectric layer, with air serving as the dielectric medium, thus yielding a low initial capacitance (~3 pF). Below 40 kPa, the gel gradually deforms and begins to contact the electrode protrusions, but some air layer remains between the ionic layer and the electrode, yielding a relatively low sensitivity of approximately 686 kPa^{-1}. In the pressure range of 40–300 kPa, the sensor enters a broad linear regime with excellent linearity ($R^2 = 0.990$; **Figure S20**). Within this range, the sensitivity peaks at 6 414 kPa^{-1} between 40 and 100 kPa, owing to the increased interfacial contact with hierarchical micro/nanostructures. As the pressure increases further to 100–300 kPa, the sensitivity gradually decreases to 3 854 kPa^{-1}, although it remains high. At ultra-high pressures (up to 800 kPa), the signal gradually saturates because of limited further compression, and the sensitivity levels off at ~994 kPa^{-1}. **Figure 3(b)** presents the average performance across multiple batches of devices, with error bars indicating the standard deviation, highlighting the good reproducibility and minimal variation. As shown in **Figure 3(c)**, our device outperforms the most recent iontronic pressure sensors by combining high sensitivity with an ultrabroad sensing range. **Table S1** is included in the supplementary information, providing a detailed comparison of recent high-performance pressure sensors. Key parameters such as sensitivity, range, linearity, and response time are highlighted, offering a comprehensive benchmark to position the advancements of our sensor within the broader field of pressure-sensing technologies.

We further investigated several key parameters that influence the sensing behavior of iontronic pressure sensors. First, the EDL effect is strongly dependent on the testing frequency of the capacitive measurement circuit. As the frequency increases from 1 kHz to 10 kHz and 100 kHz, the capacitance variation ΔC significantly decreases, while the initial capacitance C_0 remains nearly unchanged, leading to variations in both the sensitivity and detection range (**Figure S21**). Among these, 1 kHz is confirmed to yield the highest sensitivity and broadest range. Additionally, sensor stability under varying ambient temperatures and humidities is crucial for practical applications. The sensor's response under a constant load of 60 kPa was evaluated at different temperatures and humidity levels (**Figure S22(a) and (c)**). As the temperature (from −10 °C to 80 °C) and humidity (from 10% RH to 90% RH) increase, the ionic mobility increases, leading to a higher capacitance signal (**Figure S22(b) and (d)**). While some thermal drift was observed, it can be compensated for by integrating a temperature sensor for calibration[38]. However, at high temperatures, the device may lose mechanical integrity, and at low temperatures, the reduced flexibility of the hydrogel and altered water dynamics lead to early saturation. For humidity-induced drift, as shown in **Figure S23(a)**, we adopted a robust encapsulation strategy to isolate environmental moisture to facilitate practical deployment. **Figure S23(b)** shows the stability of the moisture-resistant encapsulated device under a continuous load of ~60 kPa at different environmental humidities. Moreover, the water absorption of the hydrogel also influences the sensor performance. Devices fabricated with swollen hydrogels exhibit nearly twice the capacitance signal compared with those made from dry hydrogels (**Figure S24**). Additionally, the hydrogel thickness has a modest effect on the sensing characteristics, as reflected in the $\Delta C/C_0$-pressure curves (**Figure S25**).

We also evaluated other critical performance indicators of the sensor. The limit of detection (LOD) was measured to be as low as 3.4 Pa, as shown in **Figure 3(d)**. This remarkable resolution at near-zero pressure is largely attributed to the nanoclusters anchored on the microstructured surface, which enables localized microcontact with the ionic gel even under very small loads. Moreover, when a 10 g weight is applied (**Figure 3(e)**), the device displays a rapid response and relaxation time of approximately 30 ms, surpassing the response speed of human skin (~40 ms). This rapid response is ascribed to the reduced interfacial adhesion energy induced by the hierarchical microstructures and the low surface viscosity of the dielectric layer. Furthermore, under a high-pressure load of 20 kPa, the device is capable of distinguishing a subtle pressure difference of 200 Pa, corresponding to a resolution of 1% (**Figure 3(f)**). To evaluate long-term mechanical durability, the sensor undergoes over 10 000 repeated loading/unloading cycles at 60 kPa (**Figure 3(g)**). The capacitance signal remains stable throughout, with negligible signal drift (<5%) after cycling, indicating excellent repeatability and structural robustness. Additionally, in **Figure S26**, we present SEM images taken after 10 000 cycles. The results show that the microstructures remain intact, with the majority of the nanostructures remaining intact. We further tested the device performance after 10 000 loading–unloading cycles, as shown in **Figure S27**. The results show that the device exhibits a greater nonlinear effect after long-term wear, as mechanical abrasion slightly alters the contact mechanism and compressibility. Hysteresis loop tests up to 800 kPa (four loading–unloading cycles) show minimal variation (**Figure S28**), further confirming the mechanical robustness and operational reliability of the sensor. Collectively, these superior sensing characteristics underscore the broad applicability of the proposed sensor across diverse operating conditions, including high-pressure environments—particularly for electronic skin and gesture-based interfaces, where the coexistence of high sensitivity and an ultrabroad sensing range is essential.

To validate the practical applicability of the proposed pressure sensor, we developed a multi-octave musical glove system (MoMGlove) controlled by finger gestures and multilevel pressure inputs. Five identical sensors, encapsulated with tape, were attached to the fingertip regions of the glove. The MoMGlove detects the capacitance signals from five channels to recognize finger gestures, which are subsequently mapped to corresponding musical notes (do, re, mi, fa, sol, la, and ti), as shown in **Figure 4(a)**. The thumb serves as a switch, enabling additional note combinations when pressed together with other fingers. Furthermore, the pressure input is categorized into three levels corresponding to different octaves, allowing the five-channel system to output 21 distinct notes (C3–B5) through combinations of gesture and pressure (**Figure 4(b)**). The system comprises five attached pressure sensors, two multichannel capacitive-to-digital converters (CDCs), a microcontroller serving as the central control and communication module, and a computer interface for data visualization and sound output (**Figure 4(c)**). The input signals are converted into capacitance values, which are smoothed and processed using a peak detection and classification algorithm to determine the finger gesture and pressure level. The recognized note is then rendered in a piano roll display (**Figure S29**).

To ensure ease of operation and optimal sensing performance, the pressure range is set to 0–10 N. This range is equally divided into three levels (light, medium, and heavy), and corresponding capacitance thresholds are identified to define the signal intervals, which are mapped to low, medium, and high pitch levels, respectively (**Figure 4(d)**). Notably, the capacitance variation reaches the order of μF, offering strong immunity to signal noise. A minimum threshold was also defined to prevent unintended activations when the glove was idle. In addition to pressure amplitude, the MoMGlove extracts the duration of finger pressing to determine note values such as eighth notes, quarter notes, half notes, and whole notes (**Figure 4(c)**). A demonstration melody covering three octaves was constructed to showcase the system's functionality. As illustrated in **Figure 4(f)**, the MoMGlove reliably plays a continuous melody as the channels are sequentially actuated, with a rapid, multilevel pressure response and stable output signals, demonstrating its potential as a portable wearable musical instrument. Additionally, the accuracy, repeatability, and cross-talk of the MoMGlove system were evaluated. Accuracy was assessed by having participants play 21 randomly arranged musical notes, resulting in 30 sets of data samples, with an overall accuracy of 99.1% (**Figure S30**). The small errors observed were mainly due to variations in the pressure levels applied by the subjects. Cross-talk was not observed, as the sensors are discrete and isolated. For repeatability, subjects pressed the sensors 30 times at three different pressure levels. The results in **Figure S31** show excellent repeatability across all levels. To further demonstrate the system performance, videos of repeated sensor presses at each pressure level (light, medium, and heavy) are provided, as shown in **Movies S1–S3.**

To further demonstrate the capabilities of our portable MoMGlove system, we developed a PC-based graphical user interface with multiple visualization modes for monitoring pressure signals, musical output, and interactive teaching feedback (**Figure 5(a)**). We then evaluated the system's performance in playing a musical excerpt from the score of "Castle in the Sky," which spans two octaves (**Figure 5(b)**). All notes were played using a single hand wearing the MoMGlove, without external assistance. The corresponding piano roll output shows that every note was triggered with 100% accuracy, indicating precise temporal control and note classification (**Figure 5(c)**). To assess the control of pressure and rhythm, capacitance signals from the five sensor channels were recorded during the performance (**Figure 5(d)**). The results demonstrate that after brief familiarization, users can readily adapt to the system and independently control both the pressing force and timing of individual fingers. Sharp pressure transitions during rapid pitch shifts were accurately captured in the signal curves, highlighting the sensor's high sensitivity, fast response, and short relaxation time. The wide sensing range also allows for greater tolerance to variability in finger strength, accommodating natural inconsistencies in human motion. To facilitate user training, we developed a gamified, rhythm-based interface (**Figure 5(e)**). Notes appear as colored blocks descending along five tracks, each mapped to one finger, with color encoding the required pressure level—blue for light, yellow for medium, and red for heavy. This visual-pressure mapping strategy may be further explored in immersive scenarios such as virtual reality, where multisensory interaction could enable intuitive, expressive musical experiences.

The MoMGlove also enables tactile perception and quantitative analysis of grasping actions. We evaluated this by instructing the user to grasp two objects of distinct masses—a tennis ball (~58 g) and an aluminum block (~810 g). Capacitance signals from all five channels captured the full cycle of grasping, lifting, holding, lowering, and releasing (**Figure 5(f) and (g)**). In both cases, the capacitance increased rapidly upon contact, stabilized during holding, and returned to baseline after release. Notably, the heavier block produced higher capacitance values and greater channel-to-channel variation, reflecting an uneven force distribution compared to the more uniform grip on the spherical ball. Similarly, gait analysis was performed to assess the sensor's ability to monitor pressure distribution during walking in high-pressure detection applications. A sequence of movements, including standing, shifting weight, tiptoeing, walking, and fast walking, was performed (**Figure 5(h)**). The experimental setup, shown in **Figure 5(i)**, included five sensors on the insole: toe, medial forefoot (M-FF), lateral forefoot (L-FF), arch, and heel. Capacitance pressure outputs were recorded during the movements (**Figure 5(j)**). The pressure distribution analysis revealed that the subject primarily bore weight on the toe, medial forefoot, and heel, with less pressure on the arch, indicating a high arch. Additionally, comparing the inner and outer forefoot sensors provided insight into the degree of inversion of the subject's foot.

While many previous studies on iontronic pressure sensors have focused on response curves or binary pressure detection, few have demonstrated their integration into systems that collectively reflect multiple sensing advantages. Moreover, pressure signals are rarely reused as inputs for extended functionality. In contrast, MoMGlove combines high sensitivity, wide pressure detectability, and ultrafast response within a wearable platform, enabling both expressive interaction and quantitative analysis. This integration provides a representative framework for advancing real-world applications of iontronic pressure sensing in human–machine interfaces.

Conclusion

In conclusion, we designed and fabricated a femtosecond laser-engraved iontronic pressure sensor that incorporates programmable hybrid micro/nanostructures through a single, mask-free process. The laser-induced hierarchical features synergistically enhance the interfacial area and mechanical compliance, enabling extreme sensing performance: an ultrahigh sensitivity of 6 414 kPa^{-1}, an ultrabroad sensing range up to 800 kPa, an ultralow detection limit of 3.4 Pa, a rapid temporal response, and remarkable operational stability. These results demonstrate that the conventional trade-off between sensitivity and sensing range in pressure sensors can be effectively overcome through rational micro/nanoscale structural engineering. Beyond performance metrics, the successful integration of the sensors into a wearable platform underscores their adaptability and scalability. This work highlights femtosecond laser engraving as a powerful and versatile route for manufacturing functional micro/nanostructures with extreme performance, opening avenues for next-generation electronic skins, intelligent human–machine interfaces, healthcare monitoring, and robotic applications.

Experimental section

Preparation of the ionic gel

Approximately 2 g of PVA (MW ~130 000; Shanghai Titan Scientific Co., Ltd.) was added to 18 mL of deionized water and heated at 90 °C under magnetic stirring until the polymer was completely dissolved and until a transparent viscous solution was obtained (typically after 2 h). The solution was then cooled to around 50 °C, followed by the gradual addition of 1.6 mL of H_3PO_4 (AR, ⩾85%; Shanghai Macklin Biochemical Co., Ltd.) under continuous stirring. After mixing for another 2 h, a uniform PVA-H_3PO_4 precursor gel was obtained. The mixture was poured into clean Petri dishes and maintained at room temperature (~25 °C) and 60%–65% relative humidity for one day to form solid films. Finally, the gel films were gently peeled off and cut into 7 mm × 7 mm squares, which served as the dielectric component in the pressure sensor assembly.

Fabrication of the microstructured electrode

Titanium foils with a thickness of 100 μm were polished, cleaned, and subsequently patterned with micro/nanostructures using an all-fiber femtosecond laser system (FemtoYL-100/50, YSL Photonics). The laser configuration was adjusted as follows: scanning rate of 10 mm·s^{-1}; scanning pitches of 20, 50, 100, and 200 μm depending on the desired microstructure width; and laser powers of 43, 89, 119, and 347 mW depending on the target microstructure depth. To achieve uniform morphology and sufficient depth, each pattern was etched with 10 passes. Post-etching, loose debris was removed using compressed air. Subsequently, a ~100 nm gold layer was deposited onto the micro/nanostructured Ti substrate using an electron beam evaporator (Explorer-14, Denton) to form a highly conductive Ti–Au interface. The uniformity of the gold thin film is shown in Figure S32. The Ti–Au electrodes were then cut into 7 mm × 7 mm squares, which served as the upper and lower conductive plates.

Preparation of iontronic pressure sensors

A 50-μm-thick PDMS film (Hangzhou Westru Technology Co., Ltd.) was laser-cut into square ring frames (7 mm outer and 5 mm inner dimensions) to serve as the spacing layer. This frame was laminated onto the ionic dielectric film, and the sensor stack was composed of a Ti-Au upper electrode, a PDMS spacer, a PVA-H_3PO_4 dielectric, and a Ti-Au lower electrode. The device periphery was encapsulated with PI insulating tape, and silicone sealing adhesive was used to prevent moisture ingress.

Characterization and measurements

The surface morphology and hierarchical micro/nanostructures of the laser-patterned electrodes were examined using a field-emission scanning electron microscope (FE SEM, Thermo Fisher). Mechanical loading experiments were carried out on a computer-controlled universal testing platform (ZQ-990B, Zhiqu) in conjunction with a Python program for stability testing. To measure the LOD, small paper fragments were progressively stacked onto the sensor, with the mass gradually increasing until a detectable change in capacitance was observed. The capacitance characteristics were recorded with a precision LCR meter (TH2830, Tonghui) at 1 kHz unless otherwise stated. Unless otherwise specified, all the sensors had a square shape with a 7 mm side length.

Finite element analysis

Finite element simulations were performed with the commercial Abaqus/Explicit solver. The laser-engraved Ti electrodes were represented as a linear elastic solid with a Young's modulus of 116 GPa. The ionic gel was modeled using an incompressible Mooney–Rivlin hyperelastic formulation with an effective small-strain modulus of ~100 kPa (experimentally measured; **Figure S33**) and a Poisson's ratio of 0.49. Interfacial behavior was described by hard normal contact (no penetration) combined with a Coulomb friction law, adopting a friction coefficient of 0.1 to account for nanoscale surface roughness. A uniform pressure load was applied to the upper electrode, while the lower electrode was fully constrained. The mesh employed four-node plane stress elements (CPS4) for the ionic gel and four-node reduced integration plane stress elements (CPS4R) for the electrodes. The contact area evolution was tracked during loading, and the initial contact area (A_0) at 100 Pa was extracted.

Construction of the MoMGlove system

The microcontroller unit (MCU), implemented using an Arduino Uno development board, served as the core of the data acquisition system, operating at a base clock speed of 16 MHz. Signal readout from the five MoMGlove sensing channels was achieved through two high-resolution capacitance-to-digital converter chips (FDC2214, Texas Instruments), each providing four input channels with 28-bit resolution. The internal reference clock frequency was 40 MHz, and the sensor excitation frequency was set to 10 kHz. Distinct I²C addresses were assigned to the two CDC chips. Capacitance values were acquired by the CDCs, stored in their registers, read by the MCU via the I²C interface, and then converted into capacitance units. The press detection operated with a minimum time resolution of 0.25 s, with 0.1 μF set as the minimum capacitance threshold for registering a press. Press strength was classified as light (<0.5 μF), medium (0.5–2 μF), or heavy (>2 μF), with a maximum detection limit of 3 μF. The algorithm assigned different pitches based on capacitance ranges, and the corresponding notes were output for audio-visual display.

Author contributions

Z. H. and D. S. contributed equally to this work. D. S. planned the study and supervised the project. Z. H. conceived the main experimental work. L. G., H. P., and G. Z. contributed to the device preparation and performance measurements. H. L. assisted in performing the finite element analysis. J. H., D. Z., Q. G., and R. T. analyzed the data and composed the manuscript. H. Z., H. H. Kim, X. Z., M. R., and L. Z. revised the manuscript and guided its writing. All the authors discussed the results and contributed to the writing of the paper.

Acknowledgement

This work was supported by the National Natural Science Foundation of China (52305388, BE0200030), the SJTU Explore X program, and the Shanghai Jiao Tong University Initiative Scientific Research Program (WH220402021). The authors acknowledge the provision of facilities and technical support from the Center for Advanced Electronic Materials and Devices of Shanghai Jiao Tong University. The authors also acknowledge Prof. Jing Wang from Shanghai Jiao Tong University for providing support with the super-depth-of-field camera.

Conflict of Interest

The authors declare no conflict of interest.

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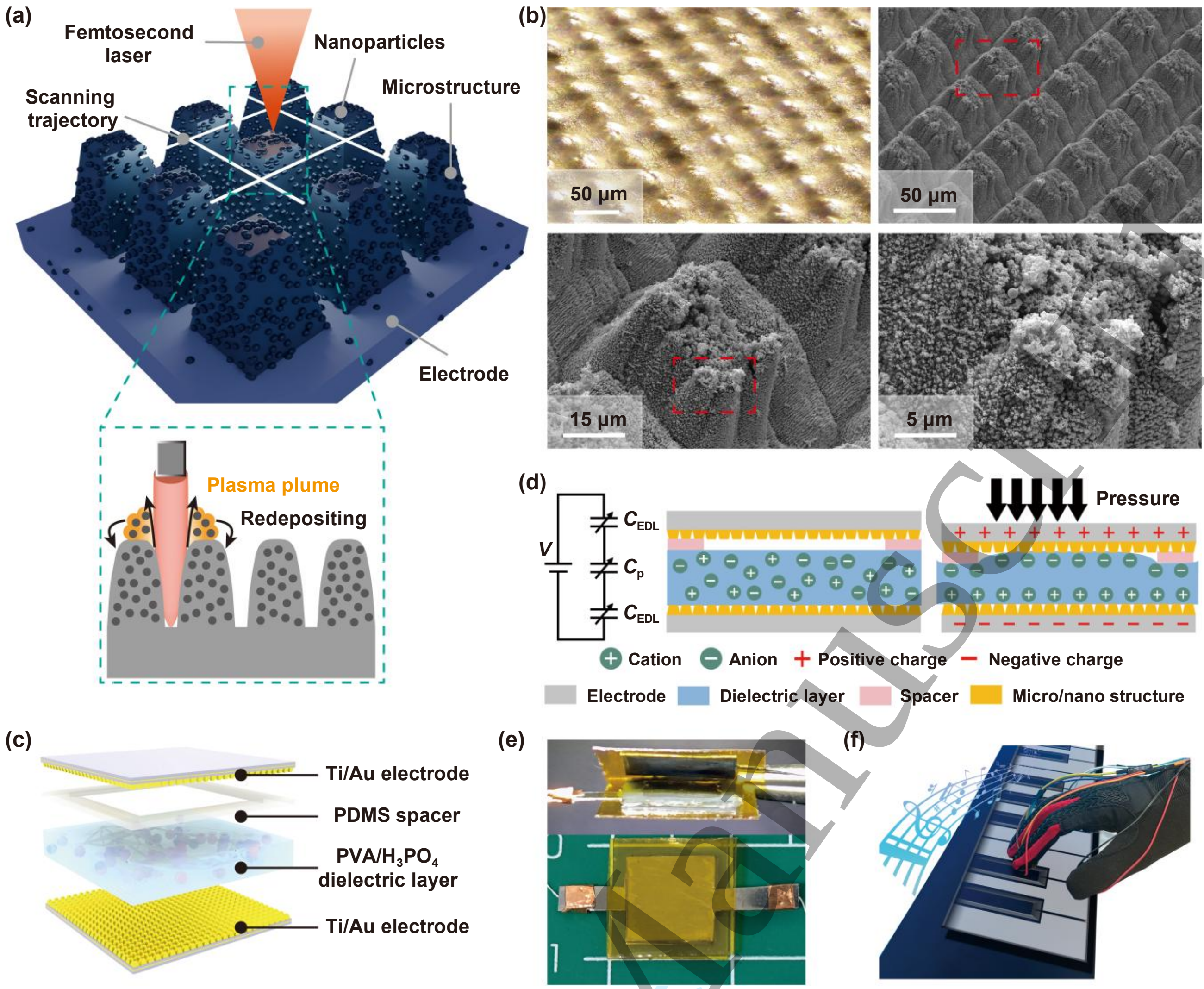

Figure 1. Design and principle of the iontronic pressure sensor with femtosecond laser-induced surface micro/nanostructures. (**a**) Fabrication process of the electrode. (**b**) Tilt-view optical and SEM images of the electrode surface microstructures at different magnifications. (**c**) Schematic illustration of the sensing mechanism of the iontronic pressure sensor. (**d**) Schematic illustration of the sensor structure based on micro/nano hybrid architecture. (**e**) Demonstration of the musical performance system developed in this study, where the sensor enables multi-level pressure sensing. (**f**) Optical photograph of the sensor integrated into the musical glove.

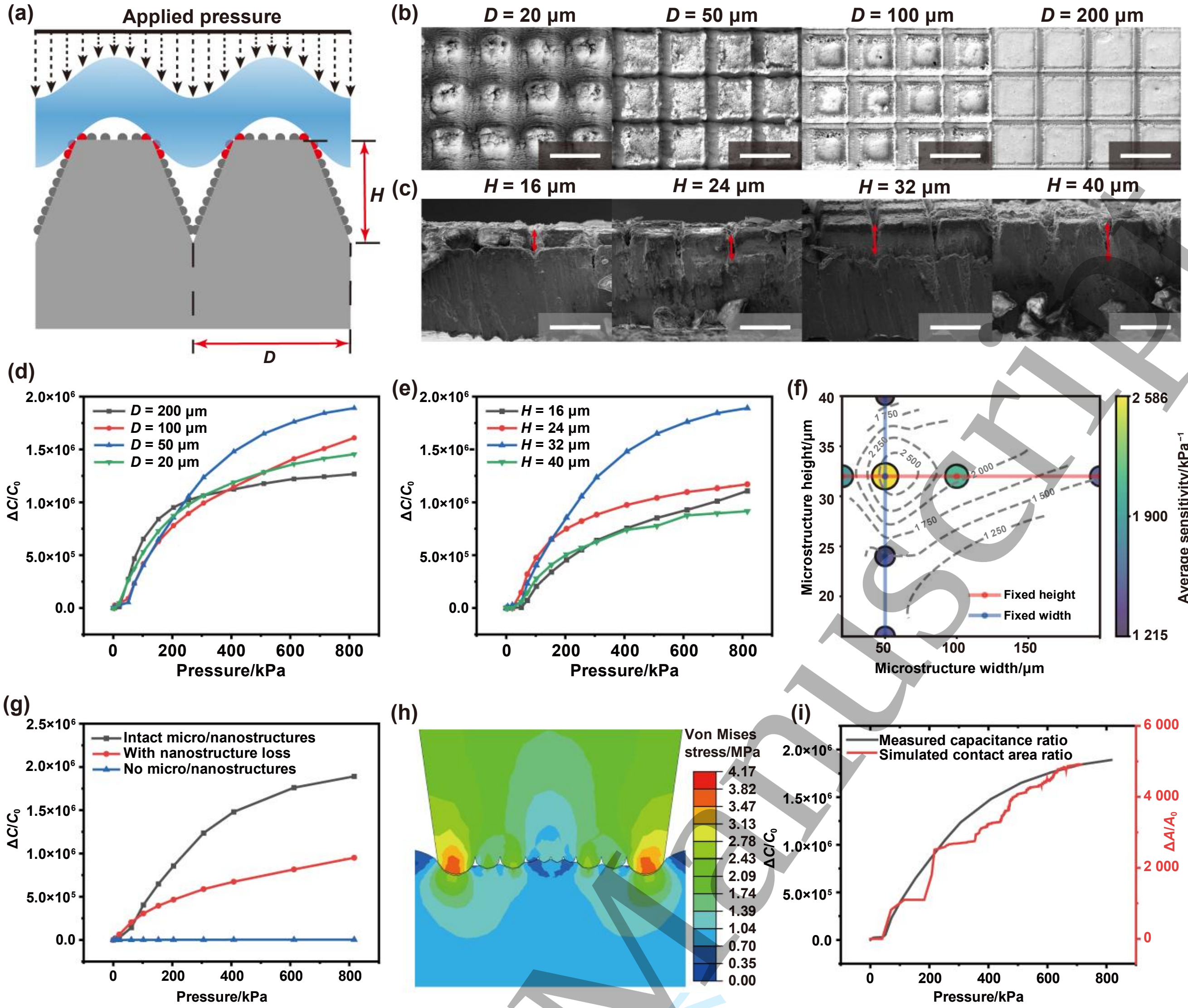

Figure 2. Characterization of micro/nanostructures and their influence on sensing performance. **(a)** Schematic cross-section of a single microstructure, showing its geometric parameters: width and height. **(b)** Top-view SEM images of microstructure arrays with varying widths (scale bars: 20 μm, 50 μm, 100 μm, and 200 μm). **(c)** Cross-sectional SEM images of microstructure arrays with different heights (scale bars: 50 μm). Comparison of pressure sensing performance for sensors with varying microstructures **(d)** widths and **(e)** heights. **(f)** Sensitivity contour map as a function of microstructure width and height under 300 kPa. **(g)** Performance comparison of pressure sensors with different micro/nanostructure configurations. **(h)** Finite element simulation results showing the stress distribution and contact region between the electrode and the dielectric layer under an applied pressure of 800 kPa. **(i)** Simulated contact area evolution as a function of pressure, overlaid with experimental capacitance data.

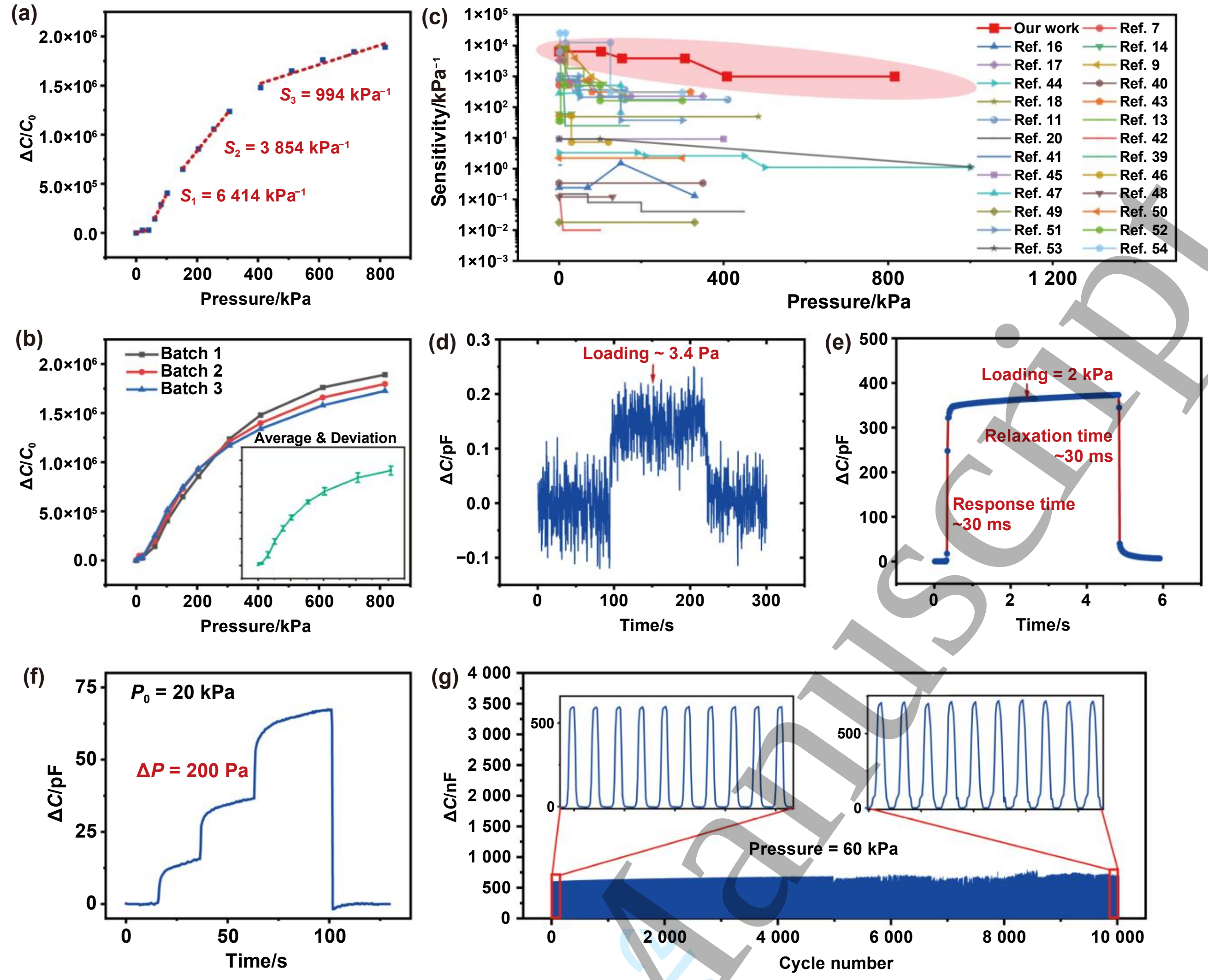

Figure 3. Sensing properties of the iontronic pressure sensor. **(a)** Normalized capacitance change over the pressure range up to 800 kPa, with a maximum sensitivity of 6 414 kPa^{-1}. **(b)** The performance curves of devices from different batches and their average performance, with error bars indicating the standard deviation. **(c)** Comparison of the sensitivity and sensing range between our sensor and previously reported iontronic pressure sensors[7,9,11,13,14,16–18,20,39-54]. **(d)** Limit of detection of the sensor. **(e)** Response and relaxation time under an applied pressure of 2 kPa. **(f)** Pressure resolution of 200 Pa under a baseline pressure of 20 kPa. **(g)** Working stability over 10 000 repeated pressure cycles under 60 kPa.

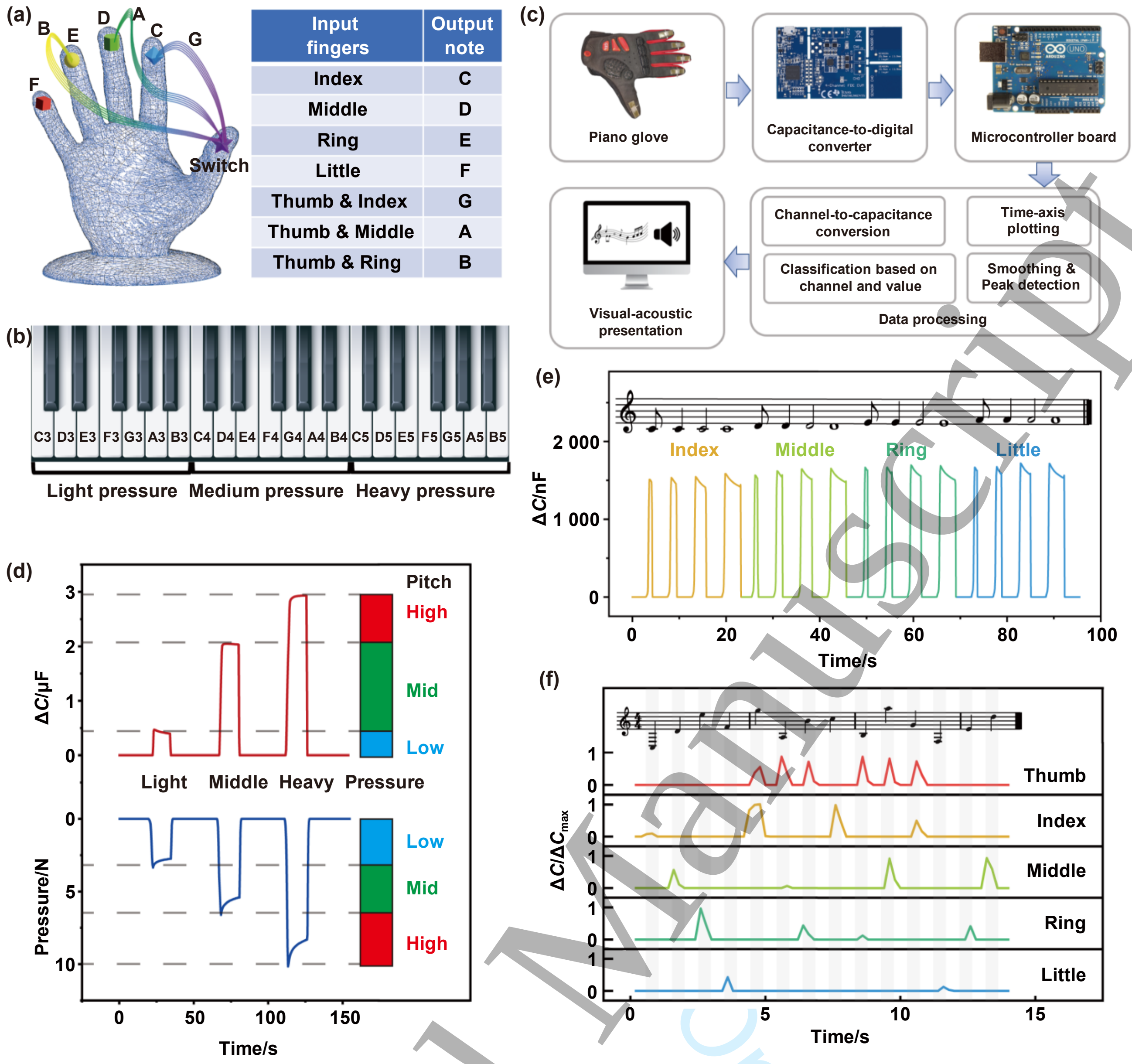

Figure 4. Application of the iontronic pressure sensor in the Multi-Octave Musical Glove (MoMGlove) controlled by finger gestures and multi-level pressure. **(a)** Diagram illustrating the correspondence between specific finger press patterns and musical note outputs. **(b)** The system enables musical note generation across three octaves (C3–B5) by combining finger gestures with three pressure levels. **(c)** Workflow of real-time signal conversion: pressure input is sensed, digitized via a CDC, processed by the microcontroller, and transformed into visual and audio outputs on a computer. **(d)** Capacitance thresholds used to distinguish three pressure levels based on measurements under equally spaced force conditions. **(e)** Pressing time determines note duration, allowing control over rhythmic values, including eighth, quarter, half, and whole notes. **(f)** Normalized capacitance changes from the five finger sensors during the performance of a melody spanning three octaves.

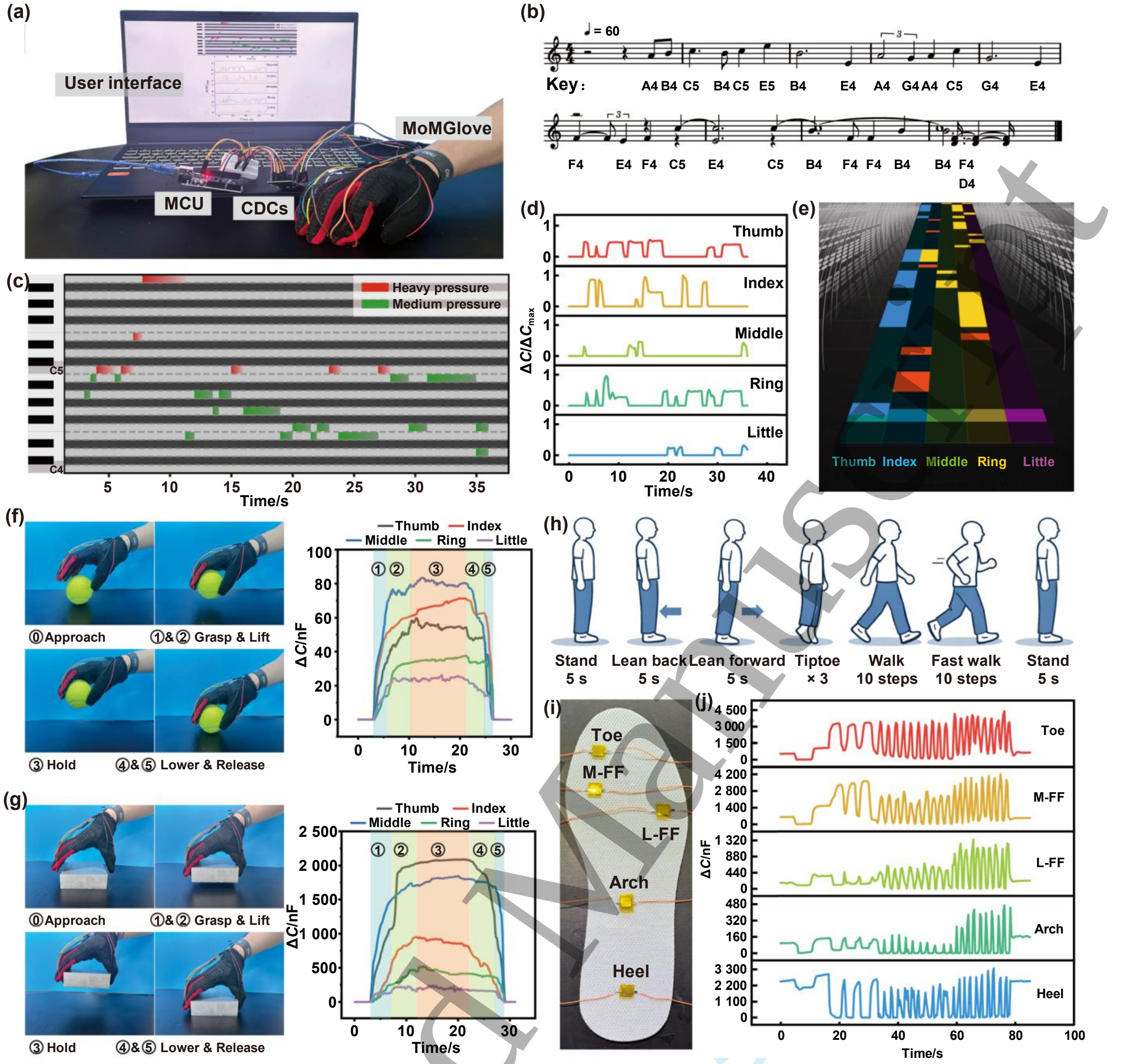

Figure 5. Interactive musical interface and human motion monitoring capability enabled by the MoMGlove system. **(a)** Structural overview of the MoMGlove system. **(b)** Excerpt from the music score of "Castle in the Sky". **(c)** Piano roll visualization confirming the successful performance of the piece "Castle in the Sky" using the MoMGlove. **(d)** Normalized capacitance variations from the five sensor channels during performance. **(e)** A rhythm-training interface, demonstrated using "Castle in the Sky", where falling colored blocks indicate which finger to press and the required pressure level—red, yellow, and blue, corresponding to high, medium, and low pressure, respectively. Sequential images and corresponding capacitance evolution curves during MoMGlove-assisted grasping, lifting, holding, lowering, and releasing of **(f)** a tennis ball and **(g)** an aluminum block. **(h)** Sequence of movements performed during the gait analysis. **(i)** Experimental setup with five sensors on the insole. **(j)** Capacitance pressure outputs recorded during the movements.